

POINTE AUX BARQUES, MI, USA

WMO: 725386

Lat: 44.022N

Lon: 82.793W

Elev: 180

StdP: 99.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.2	-13.7	-23.9	0.4	-12.5	-20.9	0.6	-11.2	13.3	-3.2	12.0	-4.0	3.7	230	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	30.2	21.9	28.3	21.3	26.7	20.6	23.7	28.0	22.7	26.6	21.8	25.4	3.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.3	26.3	21.3	16.3	25.2	20.4	15.4	24.4	71.4	28.3	67.8	26.6	64.3	25.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.2	8.1	-18.9	33.5	3.8	1.7	-21.7	34.7	-24.0	35.7	-26.1	36.6	-28.9	37.9		
			-19.4	25.5	3.6	1.2	-22.0	26.4	-24.1	27.1	-26.1	27.8	-28.7	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-4.1	-4.2	0.2	6.0	11.9	17.5	21.1	20.7	17.3	11.4	4.6	-0.5	(6)
	DBStd	10.05	5.41	5.27	5.42	4.59	4.80	4.21	3.12	2.80	3.79	4.37	4.54	4.32	(7)
	HDD10.0	1844	436	398	308	139	32	1	0	0	1	35	170	326	(8)
	HDD18.3	3827	694	631	562	370	208	66	9	8	63	221	411	584	(9)
	CDD10.0	1324	0	0	4	19	92	226	344	332	220	77	9	0	(10)
	CDD18.3	266	0	0	0	1	10	41	95	82	33	5	0	0	(11)
	CDH23.3	1488	0	0	1	11	84	274	548	389	154	26	0	0	(12)
	CDH26.7	363	0	0	0	1	22	81	139	80	36	5	0	0	(13)
(14) Wind	WSAvg	3.5	4.1	4.0	3.9	4.2	3.3	2.5	2.5	2.6	2.9	3.7	4.1	4.0	(14)
(15) Precipitation	PrecAvg	970	76	65	63	87	95	92	87	87	80	88	76	74	(15)
	PrecMax	1180	132	130	137	174	211	150	132	143	155	153	164	155	(16)
	PrecMin	810	19	24	18	33	37	44	36	44	15	19	21	27	(17)
	PrecStd	95	31	28	31	34	34	32	24	24	33	35	34	28	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	12.5	19.3	24.9	29.7	32.0	32.5	32.1	30.3	26.6	18.8	12.3	(19)
		MCWB	8.9	9.5	14.4	15.5	19.5	22.0	24.4	23.1	22.0	18.7	14.2	10.4	(20)
	2%	DB	7.9	6.8	14.6	20.7	25.8	29.1	30.0	28.8	27.3	22.8	16.0	9.1	(21)
		MCWB	5.2	3.4	10.9	13.8	17.8	21.2	22.4	22.0	20.1	17.3	11.9	6.8	(22)
	5%	DB	4.7	4.1	10.4	16.7	22.8	26.7	28.0	27.1	24.9	20.2	13.1	6.9	(23)
		MCWB	2.7	1.6	6.9	11.6	16.5	20.2	21.7	21.2	19.6	15.6	10.1	4.1	(24)
	10%	DB	2.9	2.4	7.0	12.8	20.0	24.0	26.3	25.5	22.8	17.8	11.2	4.9	(25)
		MCWB	1.1	0.4	4.2	8.9	15.1	18.9	21.0	20.0	18.8	13.9	8.5	2.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	9.1	15.0	17.0	21.6	23.6	25.2	24.8	23.6	20.8	14.8	11.0	(27)
		MCDB	11.3	11.6	18.6	22.5	26.5	29.4	30.4	28.9	28.1	24.6	17.3	12.2	(28)
	2%	WB	5.6	4.2	11.2	14.4	19.4	22.2	23.9	23.2	21.9	18.3	12.4	7.1	(29)
		MCDB	7.4	5.5	14.3	19.5	23.6	27.7	28.4	27.1	24.8	21.4	15.1	8.6	(30)
	5%	WB	2.9	1.9	7.4	11.9	17.6	20.9	22.9	22.3	20.7	16.3	10.4	5.0	(31)
		MCDB	4.2	3.8	10.5	16.5	22.1	25.3	26.6	25.8	23.5	19.4	12.8	6.5	(32)
	10%	WB	1.4	0.6	4.3	9.2	15.4	19.6	21.8	21.3	19.5	14.6	8.5	3.0	(33)
		MCDB	2.8	2.1	6.9	12.8	19.6	23.3	25.2	24.5	22.5	17.5	10.8	4.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.8	7.0	7.4	8.1	7.8	7.7	7.4	7.4	6.7	5.9	4.7	(35)
		MCDBR	9.8	10.0	11.9	14.6	12.8	11.3	10.0	9.5	10.1	10.7	9.1	7.1	(36)
	5% WB	MCWBR	8.2	8.1	8.9	9.6	7.9	6.6	5.6	5.5	5.7	6.7	7.5	6.2	(37)
		MCWBR	8.5	8.7	11.4	13.6	11.5	10.2	8.7	8.1	8.6	9.3	8.6	7.0	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.305	0.312	0.359	0.383	0.411	0.411	0.403	0.372	0.354	0.318	0.286	(40)
		taud	2.429	2.380	2.429	2.355	2.361	2.307	2.342	2.351	2.427	2.442	2.506	2.516	(41)
	Ebn at Noon	Ebn at Noon	828	888	929	905	888	861	856	850	853	819	803	817	(42)
		Edh at Noon	74	92	100	117	120	127	122	117	100	87	69	62	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.20	2.03	3.46	4.59	5.44	6.08	6.08	5.21	4.10	2.50	1.43	0.94	(44)
	RadStd	RadStd	0.12	0.23	0.36	0.40	0.47	0.37	0.26	0.24	0.39	0.25	0.20	0.10	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page